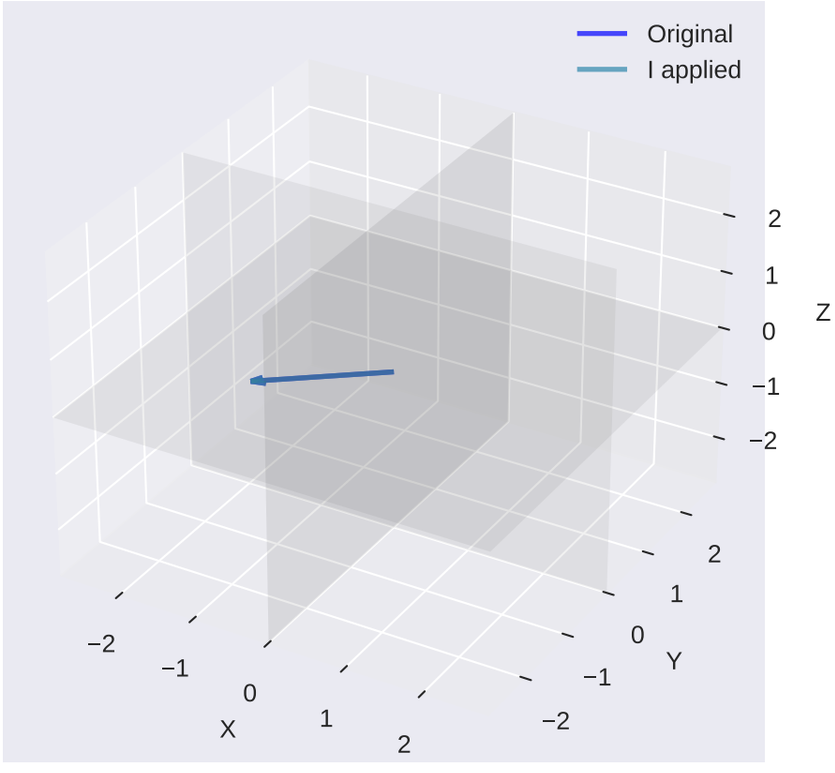
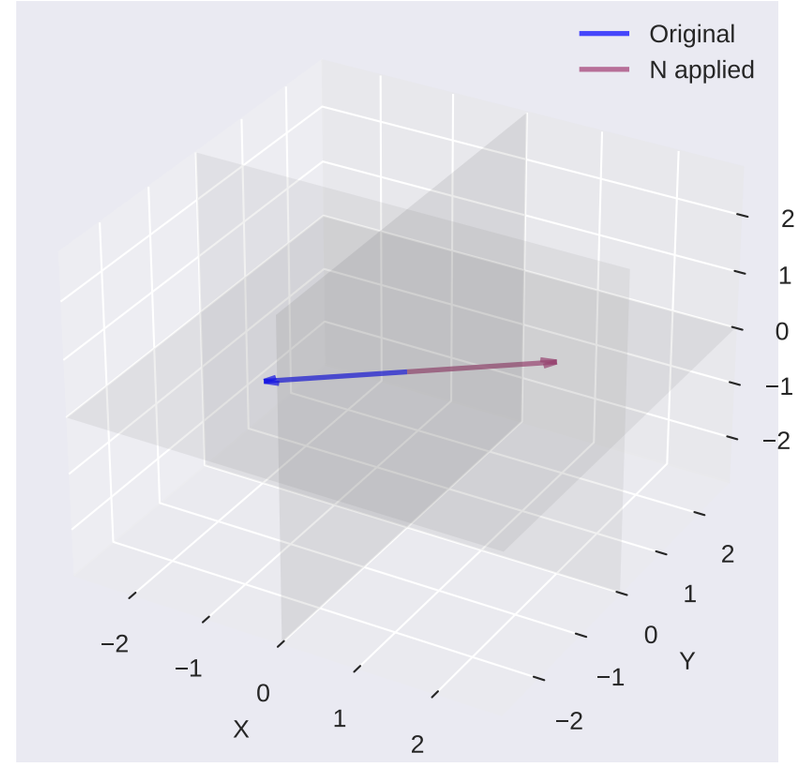
3D Vector Transformations by Klein-4 Operators
Original vector (first 3 dimensions): [-1.88371755 -0.11285862 -0.82149671]

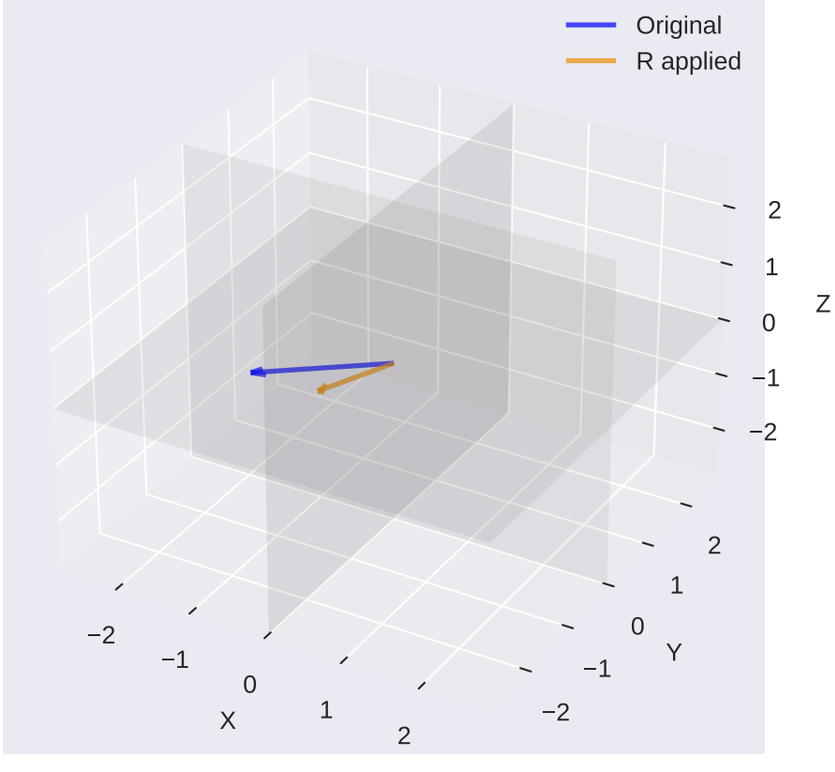
Operator I



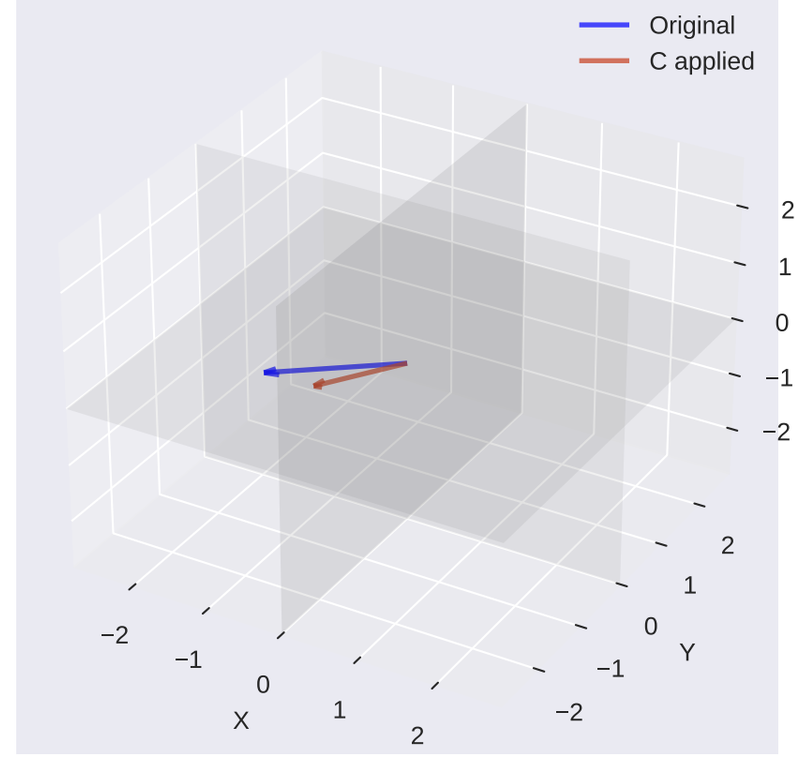
Operator N



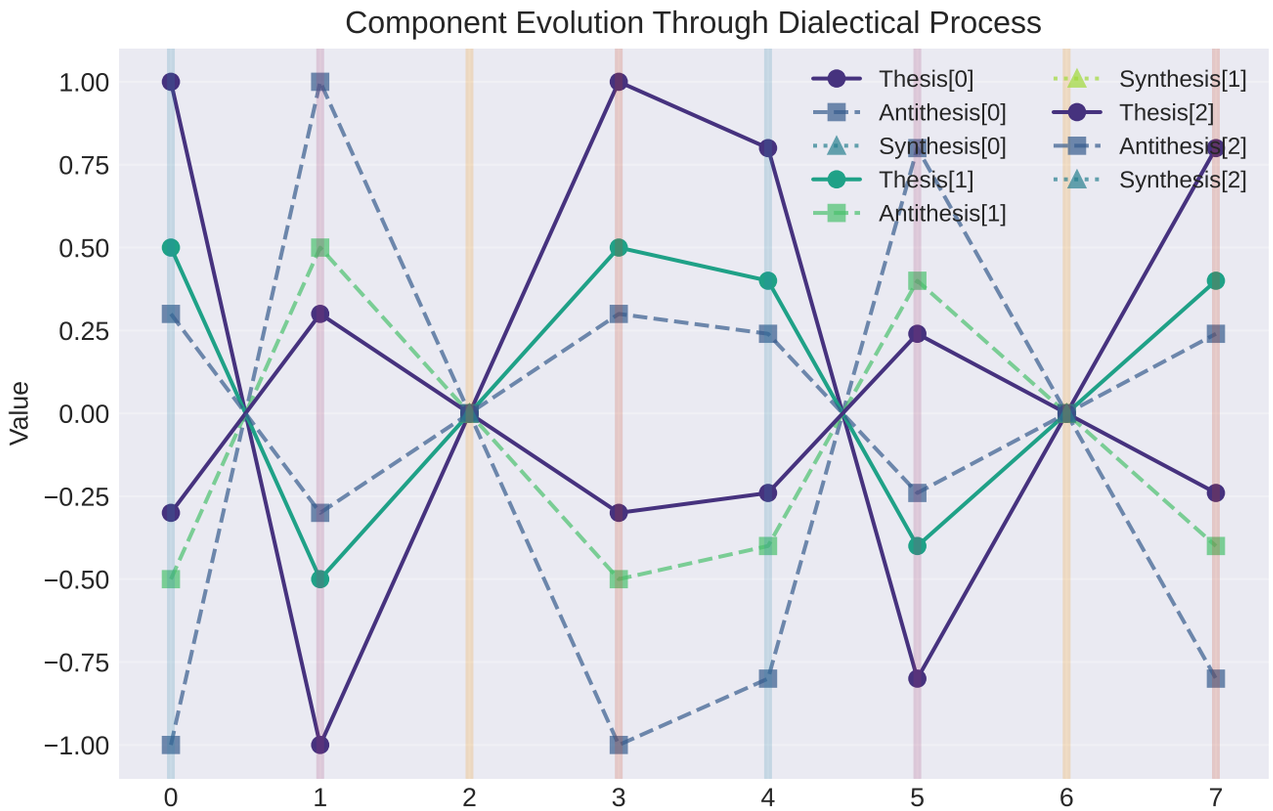
Operator R

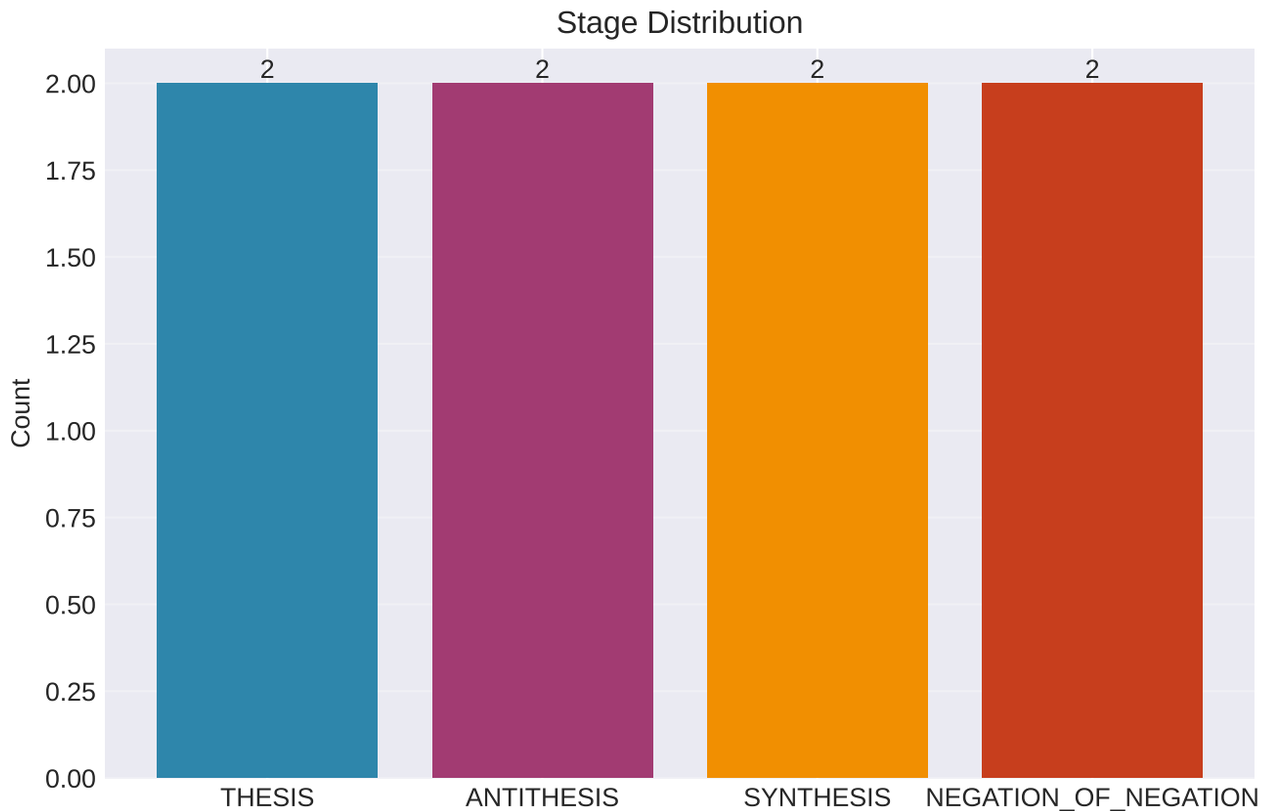


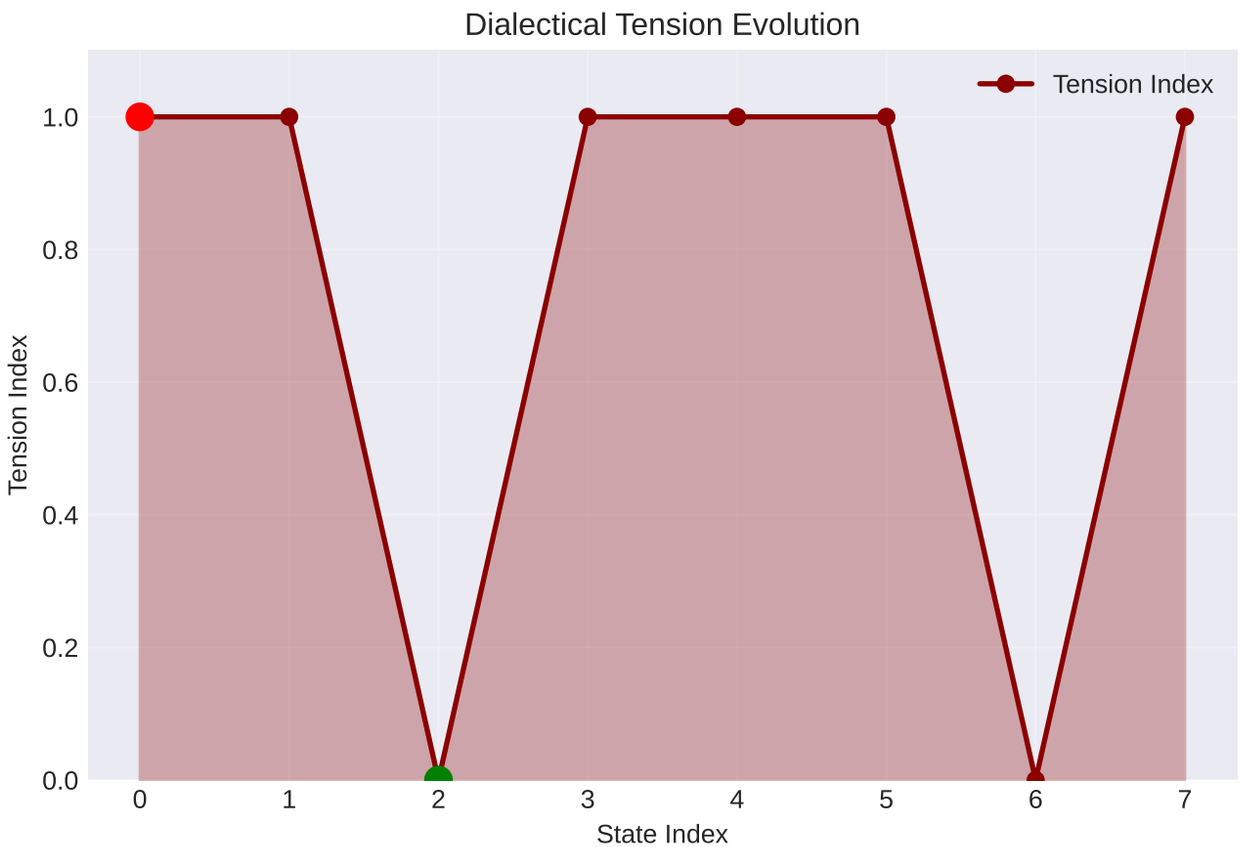
Operator C

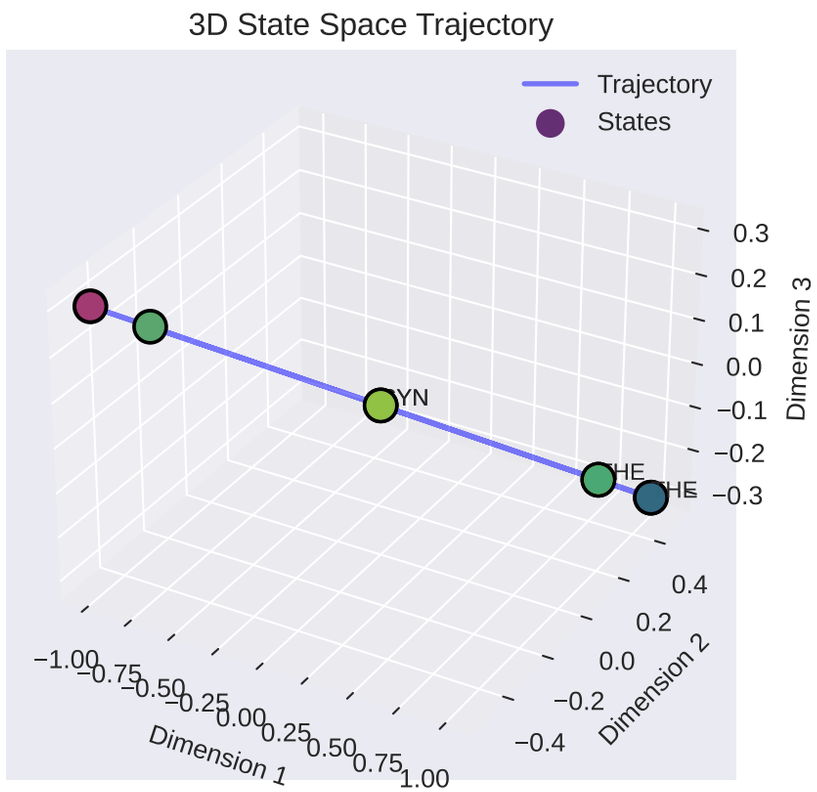


Dialectical Process Evolution
Initial thesis: [1.0, 0.5, -0.3]... | Cycles: 2 | Synthesis method: DIALECTICAL

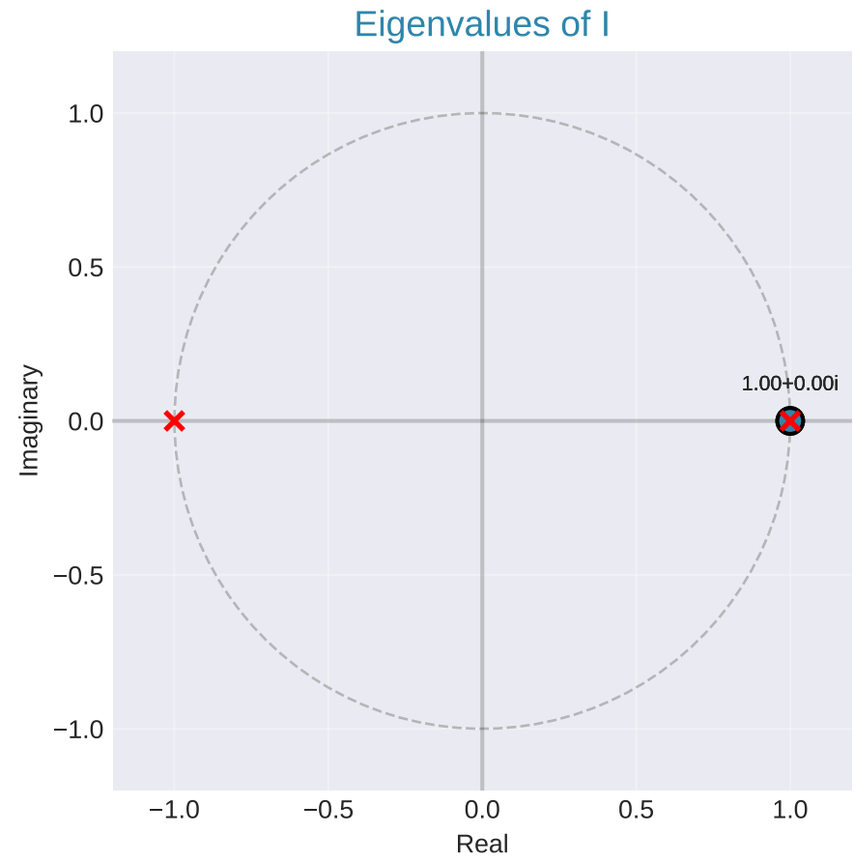


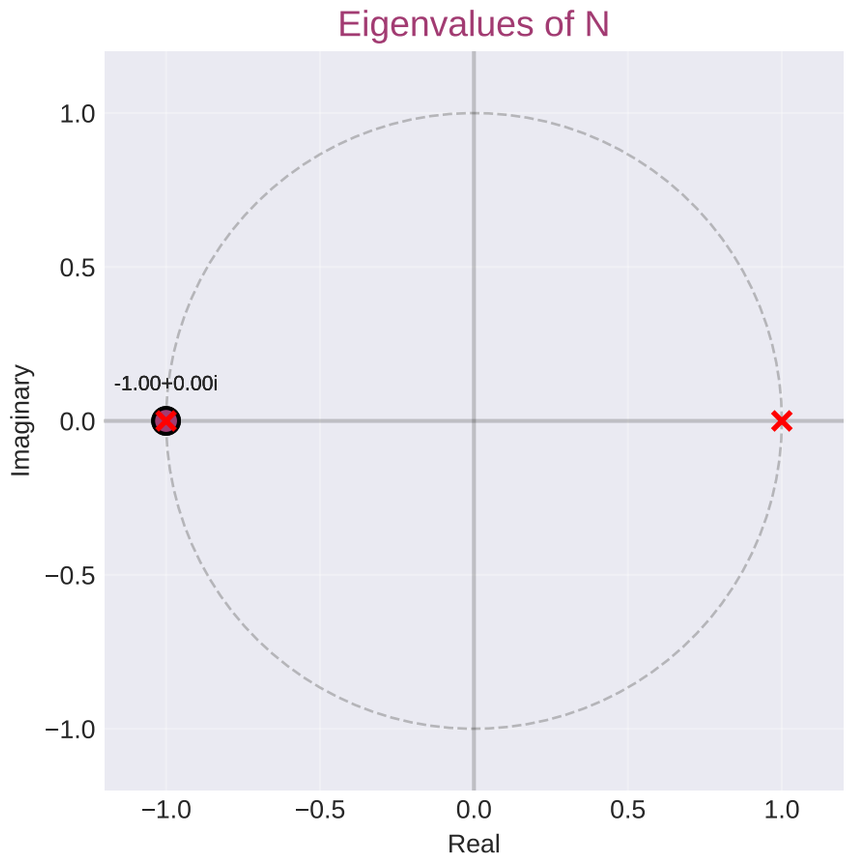


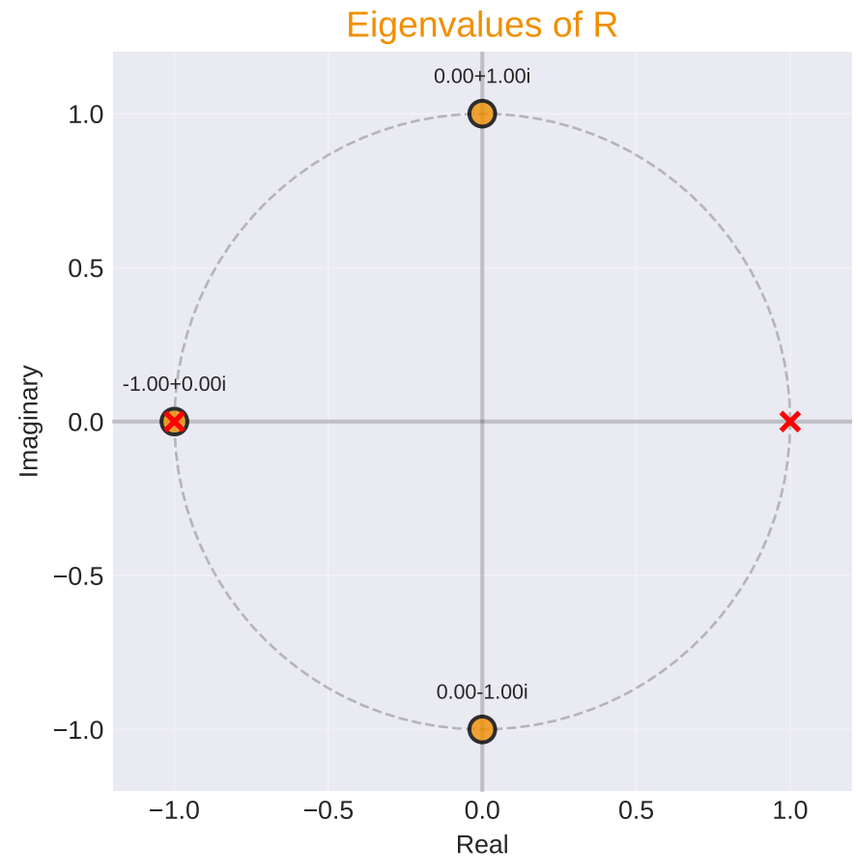


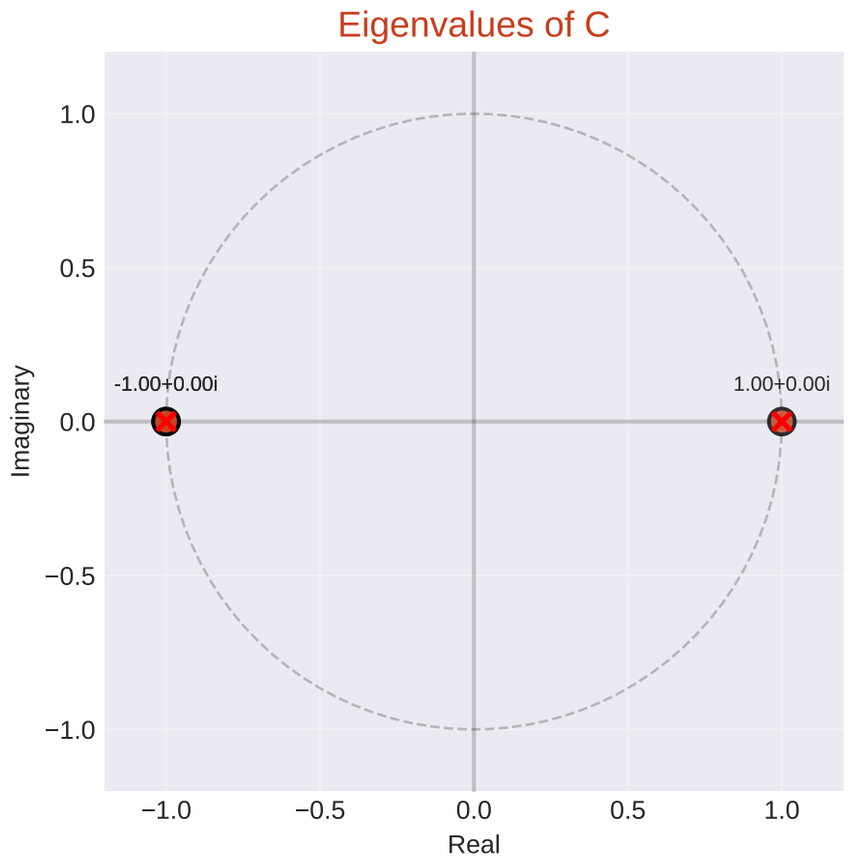


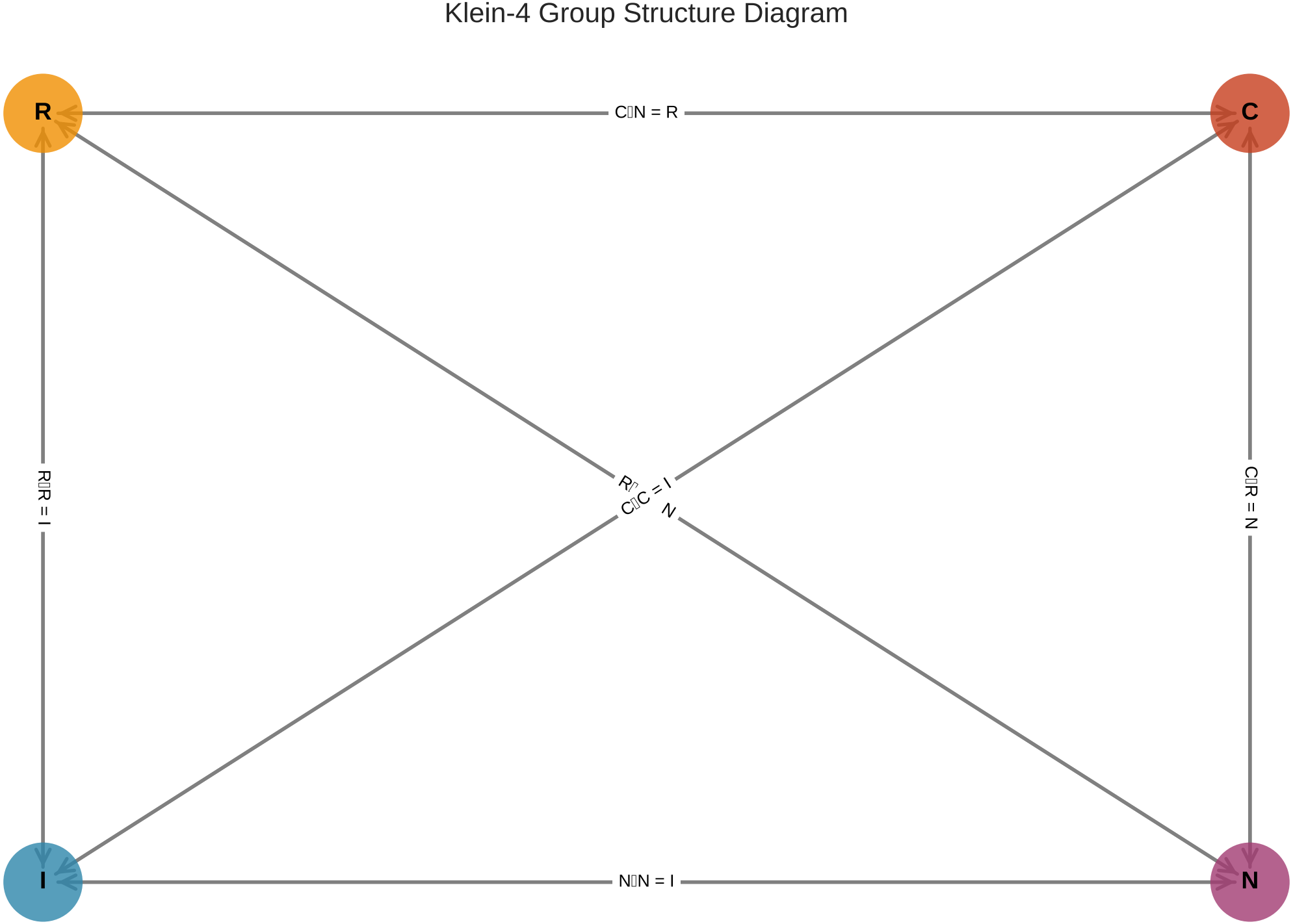
Eigenvalues on Complex Plane (Should be ±1)



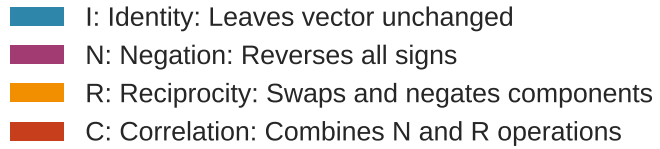






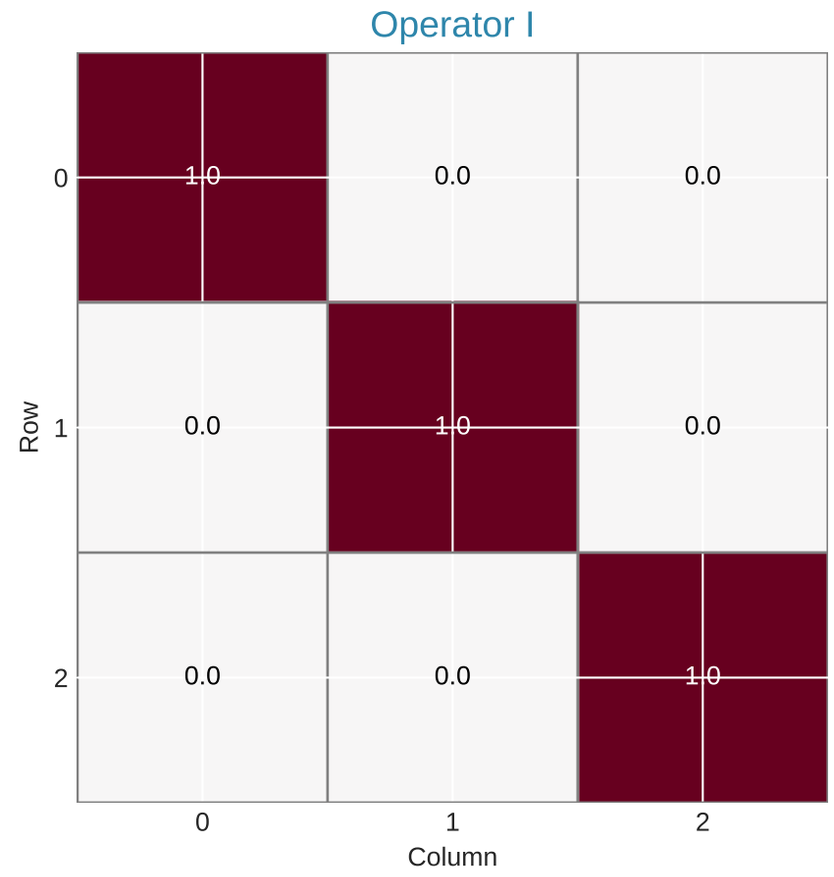

Klein-4 Group Structure Diagram
R
C
I
N
C·N = R
R·R = I
R·C = I
C·C = I
C·R = N
N·N = I

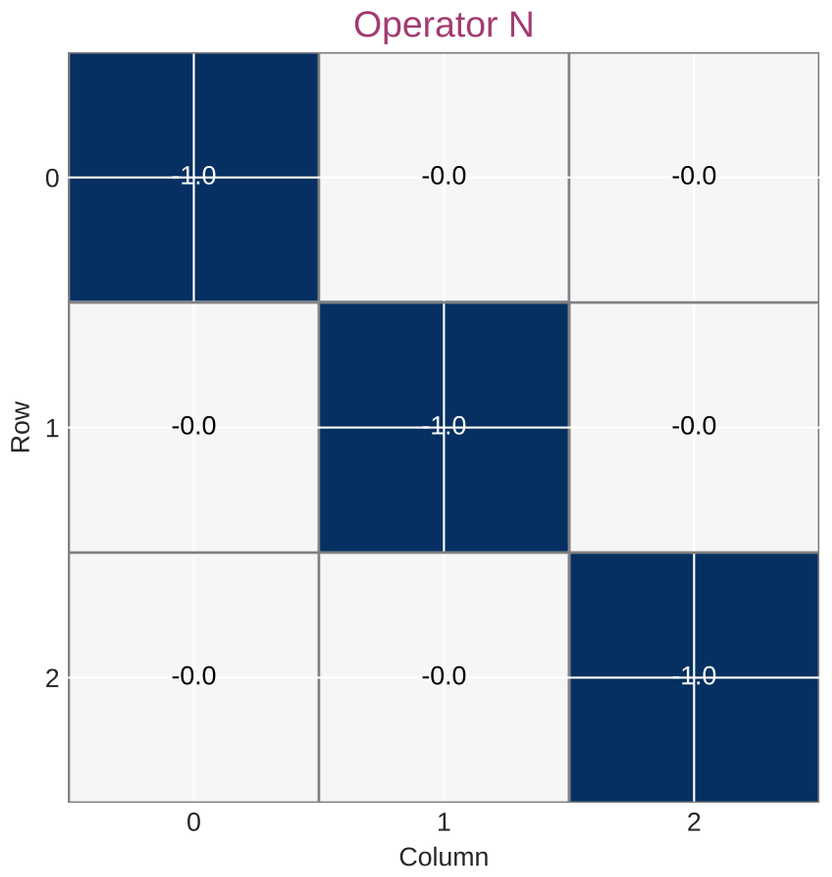
Klein-4 Group Cayley Table

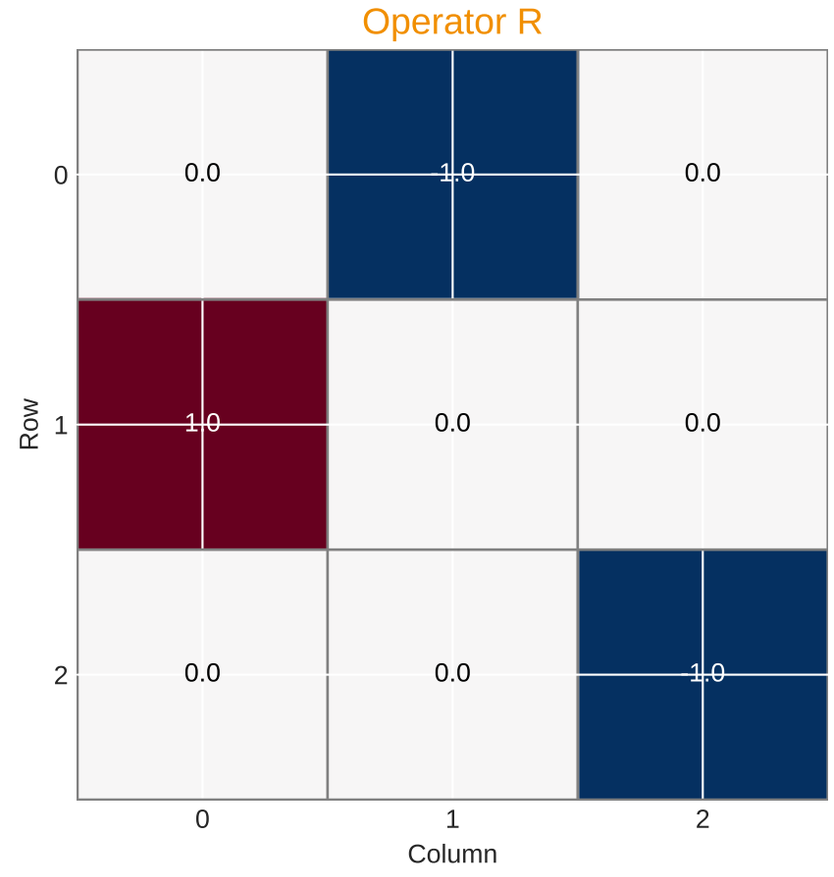


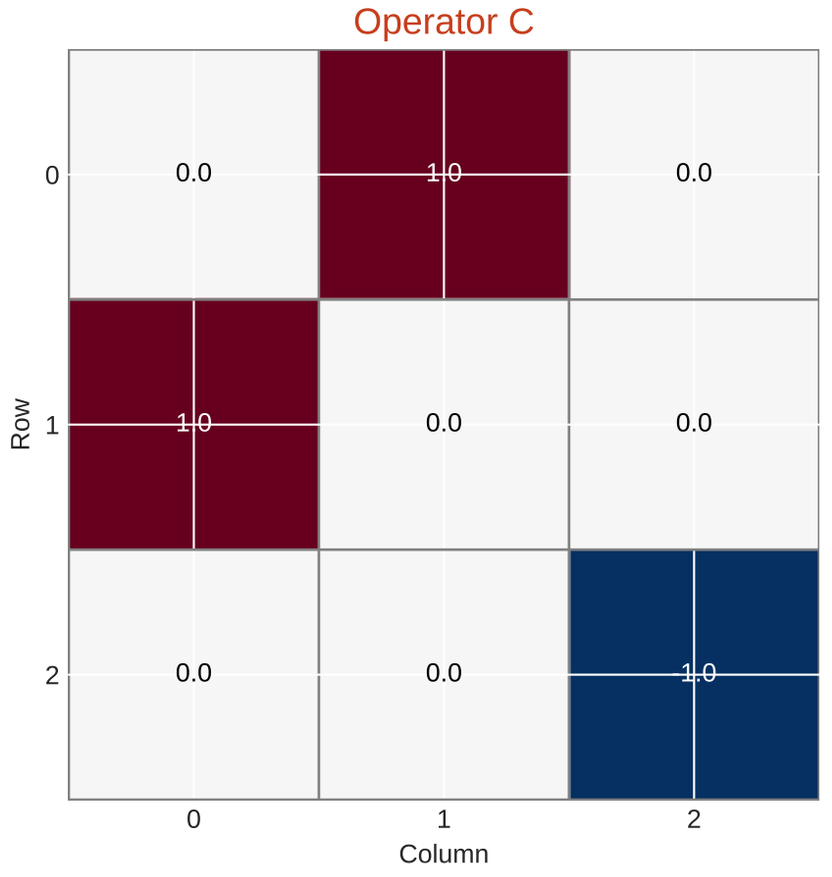
	I	**N**	**R**	**C**
I	I	N	R	C
N	N	I	C	R
R	R	C	I	N
C	C	R	N	I

Klein-4 Operator Matrices (Dimension 3)





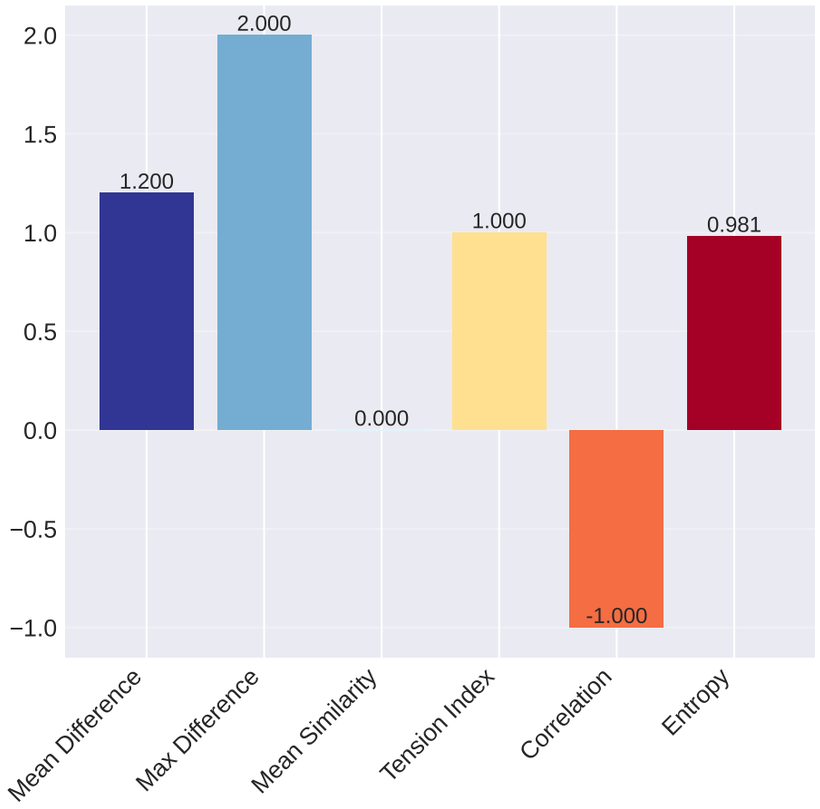




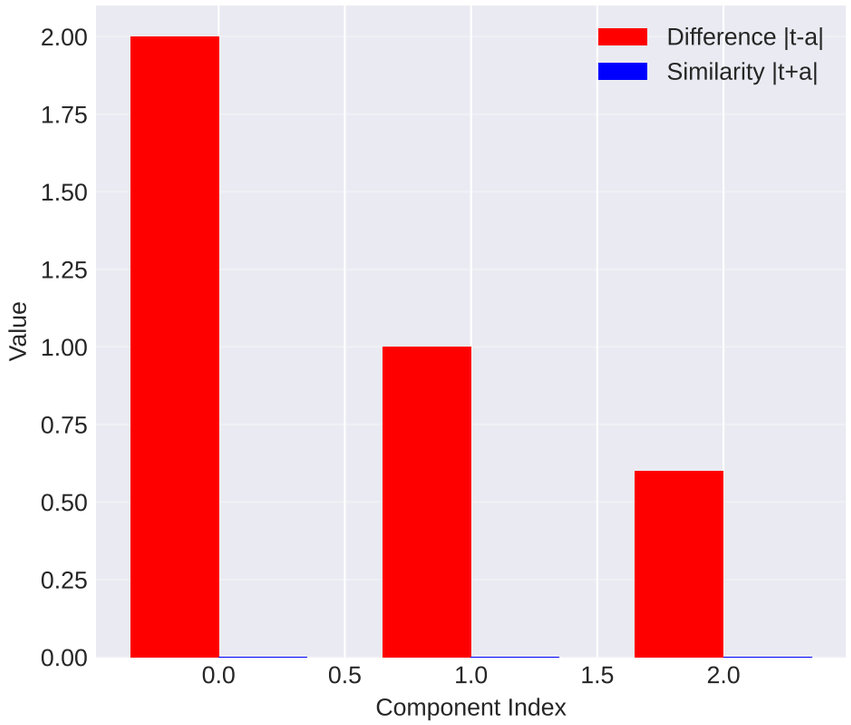
Comprehensive Klein 4 Group Property Verification
Dihedral Group D_4

Detailed Tension Analysis
Thesis: [1.0, 0.5, -0.3]... | Antithesis: [-1.0, -0.5, 0.3]...

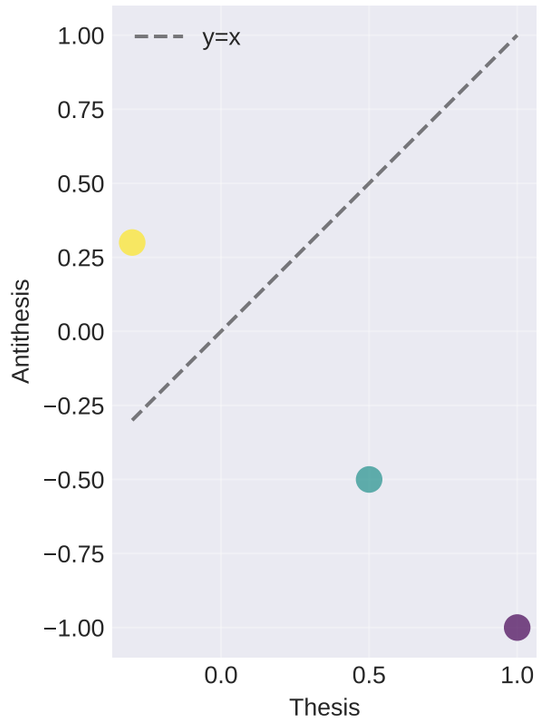
Tension Components



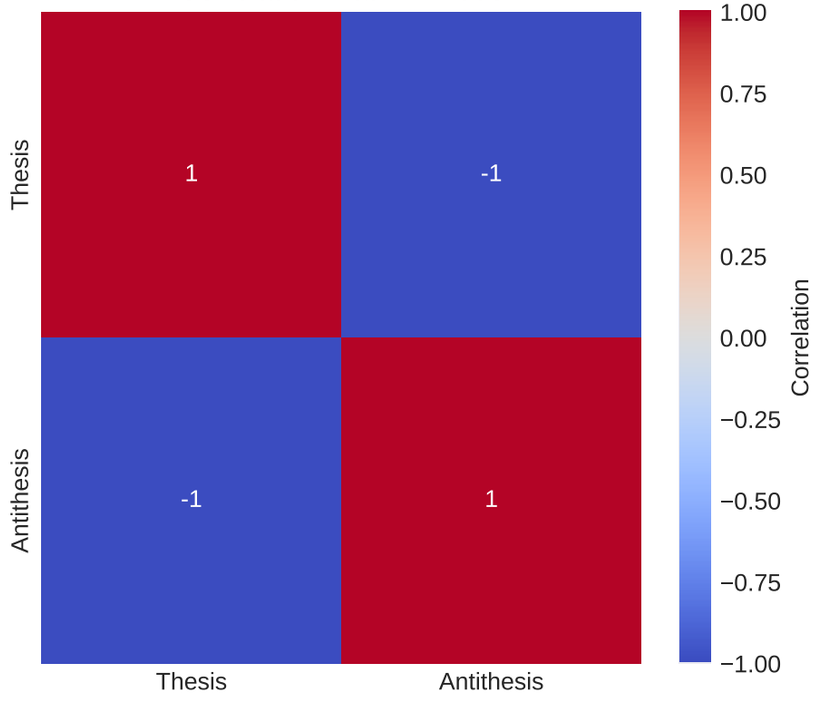
Component-wise Comparison



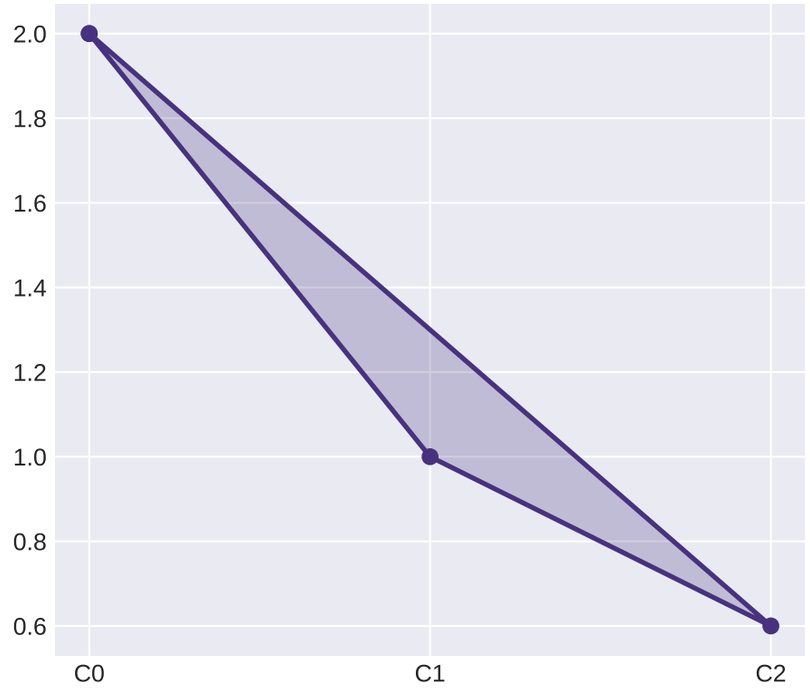
Thesis vs Antithesis



Correlation Matrix: -1.000



Radial Difference Plot



TENSION ANALYSIS SUMMARY

Stage: THESIS
Dimensions: 3
Tension Index: 1.000
Mean Difference: 1.200
Max Difference: 2.000
Correlation: -1.000
Entropy: 0.981
Has Synthesis: False

HIGH TENSION: Strong opposition